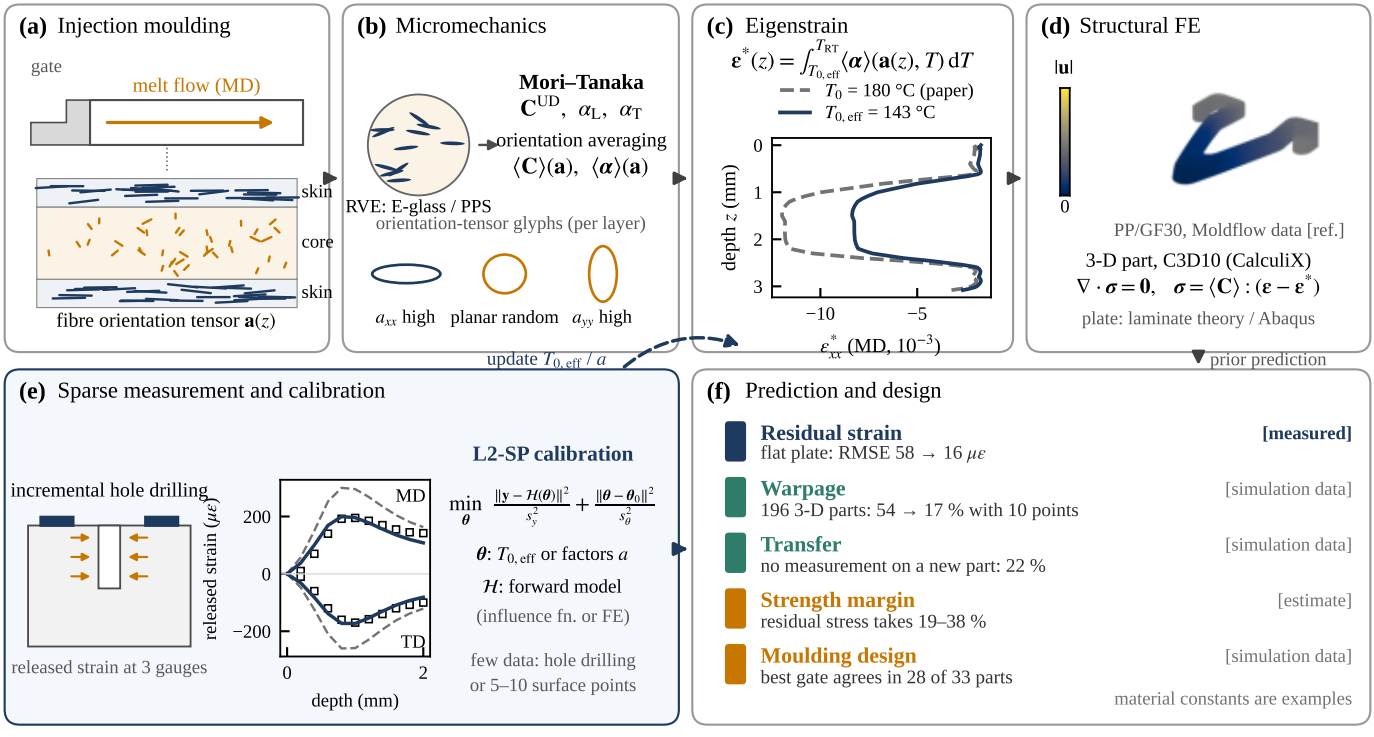

(a) Injection moulding
gate
melt flow (MD)
skin
core
skin
fibre orientation tensor a(z)
(b) Micromechanics
Mori–Tanaka
C^UD, α_L, α_T
orientation averaging
⟨C⟩(a), ⟨α⟩(a)
RVE: E-glass / PPS
orientation-tensor glyphs (per layer)
a_xx high
planar random
a_yy high
(c) Eigenstrain
ε*(z) = ∫ ⟨α⟩(a(z), T) dT
T_0 = 180 °C (paper)
T_0, eff = 143 °C
depth z (mm)
ε*_xx (MD, 10^−3)
(d) Structural FE
|u|
PP/GF30, Moldflow data [ref.]
3-D part, C3D10 (CalculiX)
∇ · σ = 0, σ = ⟨C⟩ : (ε − ε*)
plate: laminate theory / Abaqus
update T_0, eff / a
prior prediction
(e) Sparse measurement and calibration
incremental hole drilling
released strain (με)
released strain at 3 gauges
MD
TD
depth (mm)
L2-SP calibration
θ: T_0, eff or factors a
H: forward model
(influence fn. or FE)
few data: hole drilling
or 5–10 surface points
(f) Prediction and design
Residual strain
flat plate: RMSE 58 → 16 με
[measured]
Warpage
196 3-D parts: 54 → 17 % with 10 points
[simulation data]
Transfer
no measurement on a new part: 22 %
[simulation data]
Strength margin
residual stress takes 19–38 %
[estimate]
Moulding design
best gate agrees in 28 of 33 parts
[simulation data]
material constants are examples